Lecture 14

The category of finite-dimensional representations

Objectives. *Lecture 13 classified the simple modules. We now study kernels, multiplicities, tensor products and duals of all finite-dimensional modules.*

By the end of this lecture you will be able to:

- Recover a module from its multiplicity spaces by evaluation.
- Construct tensor products and duals, with equivariant evaluation and coevaluation.
- Prove $L(\lambda)^* \cong L(-w_0\lambda)$ for $\lambda \in P_+$, and compute it for $\mathfrak{sl}_3$.
- Decompose $\mathbb{C}^3 \otimes (\mathbb{C}^3)^*$ under $\mathfrak{sl}_3$ and compute its endomorphism algebra.

14.1 Kernels, cokernels and Hom spaces

Throughout, $\mathfrak{g}$ is a complex semisimple Lie algebra. Fix a Cartan subalgebra $\mathfrak{h}$, positive roots R_+, simple roots $\alpha_1, \dots, \alpha_r$ and the Weyl group W. Write

$$Q_+ = \mathbb{Z}_{\geq 0}\alpha_1 + \cdots + \mathbb{Z}_{\geq 0}\alpha_r.$$

Let C be the category of finite-dimensional $\mathfrak{g}$-modules and intertwiners. Let Vect be the category of finite-dimensional complex vector spaces and linear maps. Let $F\colon C \to \mathrm{Vect}$ be the forgetful functor, which forgets the action.

Each $\mathrm{Hom}_{\mathfrak{g}}(V,W)$ is a subspace of $\mathrm{Hom}_{\mathbb{C}}(V,W)$, and composition is bilinear. The direct sum $V \oplus W$ has action $\rho_V(x) \oplus \rho_W(x)$. Its inclusions and projections are intertwiners, so it is a biproduct in C. The zero module is a zero object.

Proposition 14.1 (Abelian structure [P]). *Let $T\colon V \to W$ be an intertwiner.*

(i) The submodule $\ker T$, with inclusion ι, is a kernel of T in C.

(ii) The quotient module $W/\mathrm{im}\,T$, with quotient map q, is a cokernel of T in C.

(iii) The map $\bar{T}\colon V/\ker T \to \mathrm{im}\,T$, $v + \ker T \mapsto Tv$, is an isomorphism of modules.

Hence C is a $\mathbb{C}$-linear abelian category, and F is faithful and exact.

Proof. (i) Lecture 1 shows that $\ker T$ is a submodule. Let $S\colon U \to V$ be an intertwiner with $TS = 0$. Then $S(U) \subseteq \ker T$, so $S = \iota S'$ for a unique linear map S'. For $x \in \mathfrak{g}$ and $u \in U$,

$$\iota S'(xu) = S(xu) = xS(u) = \iota\big(xS'(u)\big).$$

Since ι is injective, S' is an intertwiner.

(ii) Let $S\colon W \to U$ be an intertwiner with $ST = 0$. Then S vanishes on $\operatorname{im} T$, so $S = \bar{S}q$ for a unique linear map $\bar{S}$. For $x \in \mathfrak{g}$ and $w \in W$, the map $\bar{S}$ intertwines:

$$\bar{S}\big(x(w + \operatorname{im} T)\big) = S(xw) = xS(w) = x\,\bar{S}(w + \operatorname{im} T).$$

(iii) The first isomorphism theorem makes $\bar{T}$ a linear bijection. It intertwines because $T(xv) = xTv$. Applying $\bar{T}^{-1}$ to $x\bar{T}(u) = \bar{T}(xu)$ shows that $\bar{T}^{-1}$ also intertwines.

Part (iii) identifies coimage with image, the remaining axiom of an abelian category. Kernels and cokernels are the vector-space ones, so F is exact. It is faithful because morphisms in $\mathcal{C}$ are linear maps. ∎

Semisimplicity of $\mathfrak{g}$ now enters. By Weyl's theorem (Lecture 5), every object of $\mathcal{C}$ is a finite direct sum of simple objects. By Lecture 13, the simple objects are, up to isomorphism, the $L(\lambda)$ with $\lambda \in P_+$, and $L(\lambda) \cong L(\mu)$ only if $\lambda = \mu$.

14.2 The isotypic decomposition

Definition 14.2 (Multiplicity space and evaluation)**.** For V in $\mathcal{C}$ and $\lambda \in P_+$, the *multiplicity space* is

$$M_\lambda(V) = \operatorname{Hom}_{\mathfrak{g}}\big(L(\lambda), V\big).$$

Let $\mathfrak{g}$ act on $M_\lambda(V) \otimes L(\lambda)$ through the second factor:

$$x(\varphi \otimes v) = \varphi \otimes xv.$$

The *evaluation map* is

$$\varepsilon_V\colon \bigoplus_{\lambda \in P_+} M_\lambda(V) \otimes L(\lambda) \longrightarrow V, \qquad \varphi \otimes v \longmapsto \varphi(v).$$

Theorem 14.3 (Isotypic decomposition [P])**.** *Let V be a finite-dimensional $\mathfrak{g}$-module.*

(a) Each $M_\lambda(V)$ is finite-dimensional, and $M_\lambda(V) \neq 0$ for at most $\dim V$ weights λ.

(b) The evaluation map ε_V is an isomorphism of $\mathfrak{g}$-modules.

(c) In every decomposition of V into simple modules, $L(\lambda)$ occurs exactly $\dim M_\lambda(V)$ times.

Proof. Step 1: finite support. Since $M_\lambda(V) \subseteq \operatorname{Hom}_{\mathbb{C}}(L(\lambda), V)$, it is finite-dimensional. Let $\varphi \in M_\lambda(V)$ be nonzero. The kernel of φ is a proper submodule of the simple module $L(\lambda)$, hence zero. For a highest vector v_λ of $L(\lambda)$, the vector $\varphi(v_\lambda)$ is nonzero and

$$H\varphi(v_\lambda) = \varphi(Hv_\lambda) = \lambda(H)\varphi(v_\lambda) \qquad (H \in \mathfrak{h}).$$

Simultaneous eigenvectors of $\mathfrak{h}$ with distinct eigenvalue functionals are linearly independent. Hence at most $\dim V$ weights λ have $M_\lambda(V) \neq 0$.

Step 2: equivariance and naturality. The rule $(\varphi, v) \mapsto \varphi(v)$ is bilinear, so ε_V is well defined. The map ε_V intertwines because $\varphi(xv) = x\varphi(v)$. For an intertwiner $T\colon V \to V'$, let $T_*\colon M_\lambda(V) \to M_\lambda(V')$ be $\varphi \mapsto T\varphi$. Then

$$T\varepsilon_V(\varphi \otimes v) = T\varphi(v) = \varepsilon_{V'}(T_*\varphi \otimes v).$$

If T is an isomorphism, so is T_*, with inverse $(T^{-1})_*$. Then ε_V is bijective exactly when $\varepsilon_{V'}$ is.

Step 3: simple modules. Let $V = L(\mu)$. Schur's lemma gives $M_\lambda(V) = 0$ for $\lambda \neq \mu$ and $M_\mu(V) = \mathbb{C}\,\mathrm{id}$. The source of ε_V is $\mathbb{C}\,\mathrm{id} \otimes L(\mu)$, and $\varepsilon_V(\mathrm{id} \otimes v) = v$. So ε_V is bijective.

Step 4: direct sums. Let $V = V_1 \oplus V_2$ with inclusions ι_k and projections p_k. Since $\iota_1 p_1 + \iota_2 p_2 = \mathrm{id}_V$, every $\varphi \in M_\lambda(V)$ satisfies

$$\varphi = \iota_1(p_1\varphi) + \iota_2(p_2\varphi), \qquad p_k\varphi \in M_\lambda(V_k).$$

The two terms are unique, since $p_k\iota_l = \delta_{kl}\,\mathrm{id}$. Thus

$$M_\lambda(V) = \iota_{1*}M_\lambda(V_1) \oplus \iota_{2*}M_\lambda(V_2).$$

Step 2 applied to ι_k gives

$$\varepsilon_V \circ (\iota_{k*} \otimes \mathrm{id}) = \iota_k \circ \varepsilon_{V_k} \qquad (k = 1, 2).$$

The maps ι_{k*} are injective. Hence the source of ε_V splits into the images of $\iota_{1*} \otimes \mathrm{id}$ and $\iota_{2*} \otimes \mathrm{id}$. On this splitting, ε_V becomes $\varepsilon_{V_1} \oplus \varepsilon_{V_2}$ followed by the isomorphism $\iota_1 + \iota_2 \colon V_1 \oplus V_2 \to V$. If ε_{V_1} and ε_{V_2} are bijective, so is ε_V.

Step 5: conclusion. The case $V = 0$ is immediate. By Weyl's theorem and Lecture 13, $V \cong L(\mu_1) \oplus \cdots \oplus L(\mu_k)$. Steps 3 and 4, with induction on k, make the evaluation map of the right side bijective. Step 2 transfers this to V, proving (b). Steps 2–4 also give

$$M_\lambda(V) \cong \bigoplus_{j=1}^{k} M_\lambda\big(L(\mu_j)\big), \qquad \dim M_\lambda(V) = \#\{j : \mu_j = \lambda\}.$$

This is (c). ∎

The image $V[\lambda]$ of $M_\lambda(V) \otimes L(\lambda)$ under ε_V is the canonical *λ-isotypic component*. Every simple submodule isomorphic to $L(\lambda)$ is the image of some $\varphi \in M_\lambda(V)$, so lies in $V[\lambda]$. Since ε_V is an isomorphism, this component is a direct sum of copies of $L(\lambda)$. Its splittings into simple submodules correspond to decompositions of $M_\lambda(V)$ into lines, by Schur's lemma. The evaluation map has the same form as the isotypic decomposition of group representations in the finite-group course, Lecture 7.

14.3 Tensor products and duals

For modules (V, ρ_V) and (W, ρ_W) and for $x \in \mathfrak{g}$, define

$$\rho_{V\otimes W}(x) = \rho_V(x) \otimes \mathrm{id}_W + \mathrm{id}_V \otimes \rho_W(x).$$

Write $A = \rho_V(x)$, $A' = \rho_V(y)$, $B = \rho_W(x)$, $B' = \rho_W(y)$. The cross terms $A \otimes B'$ and $A' \otimes B$ occur in both products and cancel:

$$\begin{aligned}[\rho_{V\otimes W}(x), \rho_{V\otimes W}(y)] &= (AA' - A'A) \otimes \mathrm{id}_W + \mathrm{id}_V \otimes (BB' - B'B) \\ &= \rho_{V\otimes W}([x, y]).\end{aligned}$$

The *tensor unit* $\mathbf{1}$ is the trivial module $\mathbb{C}$, on which every x acts by 0.

Proposition 14.4 (Tensor structure [P]). *If $S \colon V \to V'$ and $T \colon W \to W'$ are intertwiners, so is $S \otimes T$. The associativity map $(u \otimes v) \otimes w \mapsto u \otimes (v \otimes w)$, the unit maps $c \otimes v \mapsto cv$ and*

$v \otimes c \mapsto cv$, and the flip

$$\sigma_{V,W} : V \otimes W \to W \otimes V, \qquad v \otimes w \mapsto w \otimes v,$$

are intertwiners. Moreover $\sigma_{W,V}\sigma_{V,W} = \mathrm{id}_{V\otimes W}$.

Proof. By linearity it suffices to check pure tensors. For $S \otimes T$,

$$(S \otimes T)(xv \otimes w + v \otimes xw) = xSv \otimes Tw + Sv \otimes xTw = x(Sv \otimes Tw).$$

On $(U \otimes V) \otimes W$ and $U \otimes (V \otimes W)$, x acts as the sum of three terms, each applying x to one factor. The associativity map only moves brackets. The left unit map sends $x(c \otimes v) = c \otimes xv$ to $c\,xv = x(cv)$, and the right unit map is similar. For the flip,

$$\sigma(xv \otimes w + v \otimes xw) = w \otimes xv + xw \otimes v = x(w \otimes v).$$

The identity $\sigma_{W,V}\sigma_{V,W} = \mathrm{id}$ holds for vector spaces. ∎

The associativity, unit and flip maps are bijective, so their inverses also intertwine, by the argument used for $\bar{T}^{-1}$ in Proposition 14.1. Since F is faithful, a diagram of intertwiners commutes when its image under F does. Naturality and the pentagon, triangle and hexagon identities therefore hold because they hold in Vect. Hence C is symmetric monoidal, and F is a symmetric monoidal functor with $F(V \otimes W) = F(V) \otimes F(W)$.

These names record one concrete fact. Composites of associativity, unit and flip maps that induce the same permutation of labelled factors agree, after the stated unit and associativity identifications. For example, the pentagon says that both paths of associativity maps from $((u \otimes v) \otimes w) \otimes x$ to $u \otimes (v \otimes (w \otimes x))$ send this pure tensor to the displayed tensor. The hexagon says, with associativity maps suppressed, that moving u past v and then past w by two flips gives the same result as one flip of u past $v \otimes w$.

For an operator A on V, the *dual map* is $A^{\mathsf{T}} : V^* \to V^*$, $\xi \mapsto \xi \circ A$. In a basis of V and its dual basis, the matrix of A^{T} is the transpose of the matrix of A. The *dual module* V^* has action

$$\rho_{V^*}(x)\xi = -\xi \circ \rho_V(x), \qquad D_{V^*}(x) = -D_V(x)^{\mathsf{T}}$$

in the dual basis. Transposition reverses products, so

$$\begin{aligned}[\rho_{V^*}(x), \rho_{V^*}(y)] &= \rho_V(x)^{\mathsf{T}}\rho_V(y)^{\mathsf{T}} - \rho_V(y)^{\mathsf{T}}\rho_V(x)^{\mathsf{T}} \\ &= -\big(\rho_V(x)\rho_V(y) - \rho_V(y)\rho_V(x)\big)^{\mathsf{T}} \\ &= \rho_{V^*}([x, y]).\end{aligned}$$

Lemma 14.5. *Let $\mathfrak{g}$ act on $\mathrm{End}_{\mathbb{C}}(V)$ by $x \cdot A = \rho_V(x)A - A\rho_V(x)$. The linear map*

$$\Psi : V \otimes V^* \to \mathrm{End}_{\mathbb{C}}(V), \qquad \Psi(v \otimes \xi)(u) = \xi(u)v,$$

is an isomorphism of modules.

Proof. For a basis (v_i) with dual basis (v^i), the operator $\Psi(v_i \otimes v^j)$ has the matrix unit E_{ij} as its matrix, so Ψ is bijective. For $x \in \mathfrak{g}$ and $u \in V$, the dual action gives $(x\xi)(u) = -\xi(xu)$. Hence

$$\begin{aligned}\Psi\big(x(v \otimes \xi)\big)(u) &= \xi(u)\,xv - \xi(xu)\,v \\ &= \big(\rho_V(x)\Psi(v \otimes \xi) - \Psi(v \otimes \xi)\rho_V(x)\big)(u).\end{aligned}$$

∎

Definition 14.6 (Evaluation and coevaluation)**.** With dual bases (v_i) and (v^i), define

$$\mathrm{ev}_V : V^* \otimes V \to \mathbf{1}, \quad \xi \otimes v \mapsto \xi(v); \qquad \mathrm{coev}_V : \mathbf{1} \to V \otimes V^*, \quad 1 \mapsto \sum_i v_i \otimes v^i.$$

Since $\mathrm{coev}_V(1) = \Psi^{-1}(\mathrm{id}_V)$, the coevaluation does not depend on the basis.

Proposition 14.7 (Rigidity [P])**.** *The maps* ev_V *and* coev_V *are intertwiners. With unit and associativity maps suppressed,*

$$(\mathrm{id}_V \otimes \mathrm{ev}_V)(\mathrm{coev}_V \otimes \mathrm{id}_V) = \mathrm{id}_V, \qquad (\mathrm{ev}_V \otimes \mathrm{id}_{V^*})(\mathrm{id}_{V^*} \otimes \mathrm{coev}_V) = \mathrm{id}_{V^*}.$$

Proof. Since $\mathfrak{g}$ acts by 0 on the unit $\mathbf{1}$, we need $\mathrm{ev}_V \circ x = 0$ and $x \,\mathrm{coev}_V(1) = 0$. First,

$$\mathrm{ev}_V\big(x\xi \otimes v + \xi \otimes xv\big) = -\xi(xv) + \xi(xv) = 0.$$

Second, Ψ intertwines by Lemma 14.5, and $\Psi(\mathrm{coev}_V(1)) = \mathrm{id}_V$. Hence

$$\Psi\big(x \,\mathrm{coev}_V(1)\big) = \rho_V(x)\,\mathrm{id}_V - \mathrm{id}_V\,\rho_V(x) = 0,$$

and $x \,\mathrm{coev}_V(1) = 0$ because Ψ is injective.

For the zigzags, follow $v \in V$ and $\xi \in V^*$:

$$v \mapsto \textstyle\sum_i v_i \otimes v^i \otimes v \mapsto \sum_i v^i(v)\, v_i = v,$$
$$\xi \mapsto \textstyle\sum_i \xi \otimes v_i \otimes v^i \mapsto \sum_i \xi(v_i)\, v^i = \xi.$$

Figure 14.1 draws these two identities. ∎

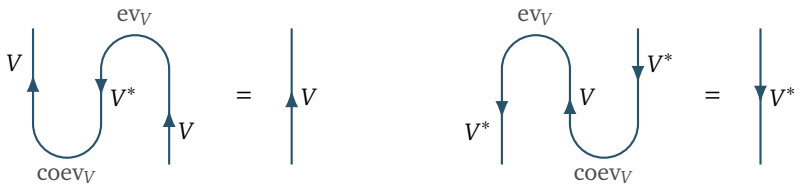


Figure 14.1: The zigzag identities, read from bottom to top. A cup is coev_V and a cap is ev_V. Each wire is labelled by its module, with arrows pointing up on V and down on V^*. Straightening a zigzag gives the identity map.

Proposition 14.7 makes V^* a left dual of V. In this convention, a left dual has its evaluation defined on $V^* \otimes V$. The module V^* is also a right dual, via the intertwiners $\mathrm{ev}_V\,\sigma_{V,V^*}$ and $\sigma_{V,V^*}\,\mathrm{coev}_V : 1 \mapsto \sum_i v^i \otimes v_i$. Their zigzags give $\sum_i v^i(v)\, v_i = v$ and $\sum_i \xi(v_i)\, v^i = \xi$.

The coproduct Δ and antipode S of Lecture 11 recover these actions on all of $U(\mathfrak{g})$:

$$u(v \otimes w) = \Delta(u)(v \otimes w), \qquad (u\xi)(v) = \xi(S(u)v) \qquad (u \in U(\mathfrak{g})).$$

These formulas agree with the actions above on generators.

14.4 The dual of a simple module

Recall that $L(\lambda)$ is the direct sum of its weight spaces, with weights in $\lambda - Q_+$ (Lecture 12). Its set of weights $\mathrm{wt}\,L(\lambda)$ is W-invariant (Lecture 13).

Lemma 14.8 (Reversing the positive roots)**.** *There is $w_0 \in W$ with $w_0(R_+) = -R_+$.*

Proof. The roots $-\alpha_1, \dots, -\alpha_r$ form a base of R, with positive roots $-R_+$. Lecture 9 shows that W acts transitively on bases, using the cited chamber theorem. So some w_0 maps the first base onto the second. Since w_0 is linear and permutes R, it maps the roots that are nonnegative combinations of the first base onto those of the second. That is, $w_0(R_+) = -R_+$. ∎

Theorem 14.9 (Dual highest weight [P])**.** *For $\lambda \in P_+$,*

$$L(\lambda)^* \cong L(-w_0\lambda).$$

Proof. Put $V = L(\lambda)$.

Step 1: weights of V^*. Choose a basis (v_k) of weight vectors, $v_k \in V_{\mu_k}$, with dual basis (ξ^k). For $H \in \mathfrak{h}$,

$$(H\xi^k)(v_l) = -\xi^k(Hv_l) = -\mu_l(H)\delta_{kl} = -\mu_k(H)\,\xi^k(v_l).$$

So $H\xi^k = -\mu_k(H)\xi^k$, and $\operatorname{wt} V^* = -\operatorname{wt} V$.

Step 2: V^* is simple. Let $U \subseteq V^*$ be a submodule. The annihilator $U^\perp \subseteq V$ of U is a submodule. Indeed, if $v \in U^\perp$ and $\xi \in U$, then $x\xi \in U$ and

$$\xi(xv) = -(x\xi)(v) = 0.$$

Since $\dim U^\perp = \dim V - \dim U$ and V is simple, $U = 0$ or $U = V^*$. By Lecture 13, $V^* \cong L(\nu)$ for some $\nu \in P_+$.

Step 3: the lowest weight of V. Let $\mu \in \operatorname{wt} V$. Then $w_0^{-1}\mu \in \operatorname{wt} V$, so $w_0^{-1}\mu = \lambda - \beta$ with $\beta \in Q_+$. Hence

$$\mu = w_0\lambda - w_0\beta \in w_0\lambda + Q_+,$$

because $-w_0$ maps R_+, and hence Q_+, into Q_+. Also $w_0\lambda \in \operatorname{wt} V$.

Step 4: comparison. By Steps 1 and 3, $\operatorname{wt} V^* \subseteq -w_0\lambda - Q_+$ and contains $-w_0\lambda$. Also $\operatorname{wt} L(\nu) \subseteq \nu - Q_+$ and contains ν. Thus $\nu + w_0\lambda$ and $-(\nu + w_0\lambda)$ both lie in Q_+. Linear independence of the simple roots gives $Q_+ \cap (-Q_+) = 0$, so $\nu = -w_0\lambda$. ∎

The proof uses only $w_0 \in W$ and $w_0(R_+) = -R_+$. Since $L(\nu)$ determines ν, the weight $-w_0\lambda$ does not depend on the choice of w_0.

For $\mathfrak{sl}_3$ with the diagonal Cartan subalgebra $\mathfrak{h}$, let $\varepsilon_i \in \mathfrak{h}^*$ be given by $\varepsilon_i(H) = a_i$ for $H = \operatorname{diag}(a_1, a_2, a_3)$, so $\varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 0$ on $\mathfrak{h}$. Then $R_+ = \{\varepsilon_i - \varepsilon_j : i < j\}$, with simple roots $\alpha_1 = \varepsilon_1 - \varepsilon_2$ and $\alpha_2 = \varepsilon_2 - \varepsilon_3$. The reflection in the root $\alpha_1 + \alpha_2 = \varepsilon_1 - \varepsilon_3$, with coroot $H_{13} = E_{11} - E_{33}$, is

$$s_{\alpha_1+\alpha_2}(\mu) = \mu - \mu(H_{13})\,(\varepsilon_1 - \varepsilon_3).$$

It exchanges ε_1 and ε_3 and fixes ε_2. Hence it sends $\varepsilon_i - \varepsilon_j$ to $\varepsilon_{4-i} - \varepsilon_{4-j}$, a negative root when $i < j$. So $w_0 = s_{\alpha_1+\alpha_2}$ satisfies $w_0(R_+) = -R_+$.

Lecture 13 identified the defining module $\mathbb{C}^3$ with $L(\omega_1)$, whose highest vector is the first standard basis vector e_1. Here $\omega_1 = \varepsilon_1$. For the other fundamental weight $\omega_2 = \varepsilon_1 + \varepsilon_2 = -\varepsilon_3$,

$$-w_0\omega_1 = -\varepsilon_3 = \omega_2, \qquad (\mathbb{C}^3)^* \cong L(\omega_2).$$

Figure 14.2 shows this reflection and the negation of weights.

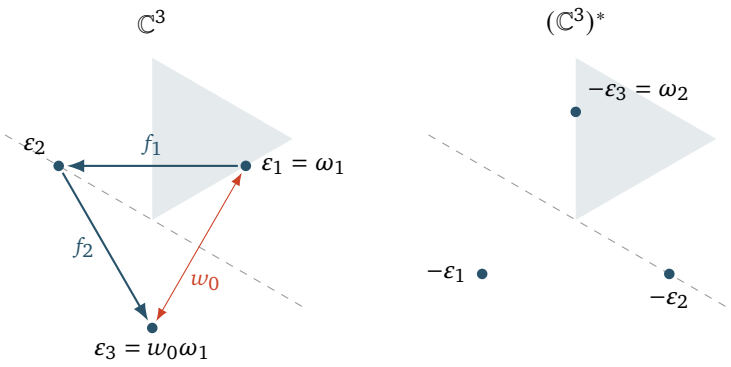


Figure 14.2: Weights of $\mathbb{C}^3$ and of its dual, with the dominant chamber shaded. On the left, f_1 and f_2 lower ε_1 to ε_3, and w_0 is the reflection in the dashed line. On the right, each weight is negated, so the new highest weight is $-w_0\omega_1 = \omega_2$.

14.5 The module $\mathbb{C}^3 \otimes (\mathbb{C}^3)^*$

Let $V = \mathbb{C}^3$ be the defining $\mathfrak{sl}_3$-module, with basis (e_i) and dual basis (e^i). The map Ψ of Lemma 14.5 identifies $V \otimes V^*$ with $\mathfrak{gl}_3(\mathbb{C})$, where $e_i \otimes e^j \mapsto E_{ij}$ and

$$x \cdot A = [x, A].$$

Under Ψ, the composite $\mathrm{ev}_V\, \sigma_{V,V^*}$ sends $v \otimes \xi$ to $\xi(v) = \operatorname{tr} \Psi(v \otimes \xi)$. So this composite is the trace. The coevaluation sends 1 to $\sum_i e_i \otimes e^i$, which Ψ sends to I_3. Hence

$$\mathrm{ev}_V\, \sigma_{V,V^*}\, \mathrm{coev}_V(1) = \sum_{i=1}^{3} e^i(e_i) = 3 = \dim V.$$

Define the *trace projector* and its complement, transported to $\mathfrak{gl}_3(\mathbb{C})$ by Ψ:

$$P = \frac{1}{3}\, \mathrm{coev}_V\, \mathrm{ev}_V\, \sigma_{V,V^*}, \qquad P(A) = \frac{\operatorname{tr} A}{3}\, I_3, \qquad Q(A) = A - \frac{\operatorname{tr} A}{3}\, I_3.$$

Both maps are intertwiners. Indeed, P is a composite of intertwiners, and $Q = \mathrm{id} - P$. The value 3 computed above gives

$$P^2 = \frac{1}{9}\, \mathrm{coev}_V\big(\mathrm{ev}_V\, \sigma\, \mathrm{coev}_V\big)\, \mathrm{ev}_V\, \sigma = \frac{3}{9}\, \mathrm{coev}_V\, \mathrm{ev}_V\, \sigma = P.$$

Figure 14.3 draws the value 3 as a closed loop and this computation as a stack of two copies of P.

Since $Q = \mathrm{id} - P$, it follows that

$$Q^2 = \mathrm{id} - 2P + P^2 = Q, \qquad PQ = QP = P - P^2 = 0.$$

So $V \otimes V^* = \operatorname{im} P \oplus \operatorname{im} Q$ as modules, with

$$\operatorname{im} P = \mathbb{C} I_3 \cong L(0), \qquad \operatorname{im} Q = \ker \operatorname{tr} = \mathfrak{sl}_3.$$

On $\mathfrak{sl}_3$ the action is the adjoint action. Submodules of the adjoint module are ideals (Lecture 1), and $\mathfrak{sl}_3$ is simple (Lecture 10). So $\operatorname{im} Q$ is simple. For $H = \operatorname{diag}(a_1, a_2, a_3)$ in $\mathfrak{h}$, the vector $e_1 \otimes e^3$, which corresponds to E_{13}, satisfies

$$\begin{aligned} [E_{12}, E_{13}] &= E_{12}E_{13} - E_{13}E_{12} = 0 - 0 = 0, \\ [E_{23}, E_{13}] &= E_{23}E_{13} - E_{13}E_{23} = 0 - 0 = 0, \\ [H, E_{13}] &= (a_1 - a_3)E_{13}. \end{aligned}$$

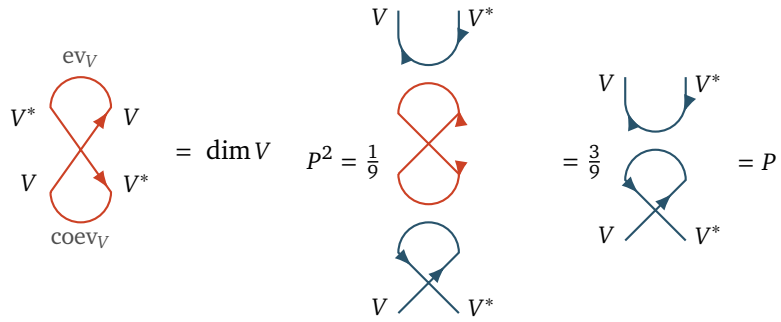


Figure 14.3: On the left, the composite $\mathrm{ev}_V \, \sigma_{V,V^*} \, \mathrm{coev}_V$ is a closed loop with value $\dim V$, read from bottom to top, and the crossing is the flip. On the right, $V = \mathbb{C}^3$. Stacking two copies of P creates one such loop, which gives the factor 3.

The root vectors E_{12} and E_{23} generate $\mathfrak{n}_+$, since $[E_{12}, E_{23}] = E_{13}$. So E_{13} is a highest vector of weight $\varepsilon_1 - \varepsilon_3 = \omega_1 + \omega_2 = \alpha_1 + \alpha_2$. Therefore

$$\mathbb{C}^3 \otimes (\mathbb{C}^3)^* \cong L(0) \oplus L(\omega_1 + \omega_2), \qquad 9 = 1 + 8.$$

Endomorphism algebra. Let $T \in \mathrm{End}_{\mathfrak{g}}(\mathfrak{gl}_3)$. With respect to $\mathfrak{gl}_3 = \mathbb{C}I_3 \oplus \mathfrak{sl}_3$, write

$$T = \begin{pmatrix} T_{00} & T_{01} \\ T_{10} & T_{11} \end{pmatrix},$$

where $T_{10} \colon \mathbb{C}I_3 \to \mathfrak{sl}_3$ and $T_{01} \colon \mathfrak{sl}_3 \to \mathbb{C}I_3$. Each block is T composed with an inclusion and with P or Q, so it is an intertwiner.

The simple modules $\mathbb{C}I_3$ and $\mathfrak{sl}_3$ have dimensions $1 \neq 8$, so Schur's lemma gives $T_{10} = T_{01} = 0$, $T_{00} = a$ and $T_{11} = b$ id for some $a, b \in \mathbb{C}$. Thus $T = aP + bQ$, and

$$\mathrm{End}_{\mathfrak{g}}\big(\mathbb{C}^3 \otimes (\mathbb{C}^3)^*\big) = \mathbb{C}P \oplus \mathbb{C}Q \cong \mathbb{C} \times \mathbb{C}.$$

Both isotypic components are simple. By the description of isotypic components, $\mathbb{C}I_3 \oplus \mathfrak{sl}_3$ is the unique decomposition into simple submodules.

14.6 The structure of the category

Kernels, quotients, sums, tensor products and duals carry the module structures given above. The next terminology records their compatibility. In this terminology, abelian records kernels and cokernels, symmetric monoidal records the tensor product and the flip, and rigid records the duals with evaluation and coevaluation. Part II uses these constructions. It does not require memorising the names of the coherence diagrams.

A *rigid symmetric tensor category* is a $\mathbb{C}$-linear abelian symmetric monoidal category with finite-dimensional Hom spaces and objects of finite length, in which $(S, T) \mapsto S \otimes T$ is bilinear, $\mathrm{End}(\mathbf{1}) = \mathbb{C}$, and every object has a left and a right dual.

Theorem 14.10 (Structure of C). *Let $\mathfrak{g}$ be complex semisimple. Then C is a $\mathbb{C}$-linear semisimple abelian rigid symmetric tensor category with $\mathrm{End}_{\mathfrak{g}}(\mathbf{1}) = \mathbb{C}$. The functor F is faithful, exact and tensor. If $\mathfrak{g} \neq 0$, then C has infinitely many isomorphism classes of simple objects.*

Proof. Propositions 14.1, 14.4 and 14.7 and the right duals give the stated structures. Weyl's theorem gives semisimplicity. Hom spaces are finite-dimensional. Dimensions bound the length of strict chains of submodules. Bilinearity of $(S,T) \mapsto S \otimes T$ holds for linear maps, hence for intertwiners. Every linear map $\mathbb{C} \to \mathbb{C}$ intertwines, so $\mathrm{End}_{\mathfrak{g}}(\mathbf{1}) = \mathbb{C}$. If $\mathfrak{g} \neq 0$, then $R \neq \emptyset$ and $r \geq 1$. The weights $m\omega_1$, $m \geq 0$, are distinct and dominant, so the modules $L(m\omega_1)$ are pairwise nonisomorphic. ∎

If the zero algebra is admitted as semisimple, its category is Vect, with one simple object $\mathbb{C}$.

Part II studies the multiplicities $\dim M_\nu(L(\lambda) \otimes L(\mu))$ of simple modules in tensor products.

14.7 Part I in retrospect

Part I proved two classifications. Lectures 7–10 attach a connected Dynkin diagram to a complex simple Lie algebra. Such algebras correspond to the diagrams of types A to G, using the imported results named in Lecture 10. Lectures 11–13 show that the finite-dimensional simple modules of a complex semisimple Lie algebra correspond to the weights in P_+. Weyl's theorem in Lecture 5 makes every finite-dimensional module a direct sum of these. This lecture adds tensor products and duals.

These results name each simple module, but they do not yet compute it. Fix λ, μ and ν in P_+. We have no general method yet for three numbers:

$$\dim L(\lambda), \qquad \dim L(\lambda)_\mu, \qquad \dim M_\nu\big(L(\lambda) \otimes L(\mu)\big).$$

For $\mathfrak{sl}_2$, Lecture 6 gives all three. Beyond $\mathfrak{sl}_2$ we have only hand calculations such as §14.5. Figure 13.2 of Lecture 13 printed the multiplicities of $L(2\omega_1 + 2\omega_2)$ without a method.

Part II answers this computational question. Lecture 16 proves Weyl's character formula, using the characters of Lecture 15. Lecture 17 derives from it the dimension of $L(\lambda)$ and each weight multiplicity. Lecture 17 also decomposes a given module, and Lecture 19 computes tensor-product multiplicities in type A.

Self-checks

(1) Identify $L(2\omega_1 + \omega_2)^*$ for $\mathfrak{sl}_3$.

(2) Let $V = L(\lambda) \oplus L(\lambda) \oplus L(\mu)$ with $\lambda \neq \mu$. Find $\dim M_\lambda(V)$. Is the splitting of $V[\lambda]$ canonical?

(3) Why does the dual action carry a minus sign?

Answers. (1) $-w_0\omega_1 = \omega_2$ and $-w_0\omega_2 = w_0\varepsilon_3 = \varepsilon_1 = \omega_1$, so $L(2\omega_1 + \omega_2)^* \cong L(\omega_1 + 2\omega_2)$. (2) Two. No: the diagonal $\{(v, v, 0)\}$ is another simple submodule of $V[\lambda]$. (3) Transposition reverses brackets. The sign restores the bracket rule and makes ev_V equivariant.

Summary

Let $\mathfrak{g}$ be complex semisimple. Evaluation identifies each finite-dimensional module V with $\bigoplus_\lambda M_\lambda(V) \otimes L(\lambda)$, with finitely many nonzero terms. Evaluation, coevaluation and the flip are intertwiners. Hence C is a semisimple rigid symmetric tensor category. For $\lambda \in P_+$, the dual of $L(\lambda)$ is $L(-w_0\lambda)$.

Reading

A. Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras*, Sections 4.2 and 8.1. P. Etingof, *Lie Groups and Lie Algebras* (arXiv:2201.09397), the sections on tensor products and dual representations. J. E. Humphreys, *Introduction to Lie Algebras and Representation Theory*, Section 10.3, for the chamber theorem behind the transitivity of W on bases.